Mechanism Efficiency Showdown

Compete to transfer a fixed payload through a required path using the least input energy or force.

Advanced	2 class periods	Competition

Engineering Context

A ground-support team needs a mechanism to move a delicate payload between two stations. Multiple teams will propose machines and compare efficiency, reliability, and complexity.

What You Will Practice

- Combine simple machines
- Measure input force and distance
- Calculate work and efficiency
- Defend a design using performance evidence

What You Need

- Approved pulleys, gears, levers, wheels, axles, or linkages
- Spring scale
- Meter tape
- Fixed payload
- Engineering notebook

Student Instructions

1. Review the payload path and boundaries.
2. Sketch two mechanism concepts that use at least two simple-machine elements.
3. Estimate mechanical advantage and input travel.
4. Build one design and complete a low-load functional test.
5. Measure average input force and input distance for three official trials.
6. Calculate input work and compare it with useful output work.
7. Record success rate and completion time.
8. Prepare a one-minute defense of the design.

Engineering Requirements

- Move the payload from start to finish without direct hand contact
- Use at least two mechanical elements
- Complete three official trials
- Provide work and efficiency calculations
- Scoring includes efficiency, reliability, and simplicity

Constraints & Safety

- No launching or free-falling payloads
- Use only approved stored energy
- Inspect supports and connections before testing
- Stop immediately if parts bind or fail

What You Submit

- Two concept sketches
- Calculations
- Official trial table
- Efficiency result
- Design defense

Reflection Questions

1. What feature improved efficiency most?
2. Did the most efficient design also finish fastest?
3. How did part count affect reliability?
4. What would you change for a more fragile payload?

Engineering Tip Measure force while the system moves at steady speed; peak starting force may not represent the full trial.	**Common Mistake** Comparing only completion time ignores input effort and reliability.
Stretch Goal Redesign to complete the same task with 25% fewer parts.	**Career Connection** Mechanical design and operations engineers evaluate cargo-handling systems using efficiency, reliability, maintainability, and safety.